

BERNAT FONTANET ALBINYANA

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Profile

Software and Controls Engineer with a background in Telecommunications Engineering and currently pursuing a Master's degree in Deep Learning at UPC Barcelona.

Hands-on experience in Formula Student EV & DV development, focused in vehicle controls, embedded systems, telemetry, autonomous systems and EV powertrain technologies.

Seeking opportunities within a high-performance motorsport engineering environment.

Experience

Software Developer — NXP TrustMotion (formerly TTTech Auto)

Sept 2024 – Present | Barcelona, Spain

- Development of custom central gateway ECU solutions for Software Defined Vehicle (SDV) architectures for premium OEMs.
- Contribution to real-time automotive software systems following AUTOSAR standards.
- Integration of advanced connectivity, communication, and network security functionalities.
- Responsible for ECU diagnostics implementation and validation activities.
- Development of network routing and embedded software integration features.
- Technical coordination across three engineering projects within an Agile environment.

Technologies

AUTOSAR, Embedded C, CAN bus communication, Ethernet communication, ECU diagnostics, network routing, software integration, real-time systems

Vehicle Controls Engineer — BCN eMotorsport Formula Student Team

Sept 2023 – May 2025 | Barcelona, Spain

- Integration of torque vectoring and traction control systems for a 4WD electric and driverless race car, including real-time C++/ROS deployment and track validation.
- Enhancement of telemetry acquisition, signal processing and data analysis.
- Accelerometer and GPS/GNSS sensor integration, setup, calibration, and validation.
- Optimisation of ROS-based autonomous driving pipeline, including perception and lap-time.
- Integration with EV powertrain systems, inverters and PMSM motor control.
- Applied IPG CarMaker for vehicle dynamics analysis and control systems simulation.
- Collaboration within a multidisciplinary motorsport engineering environment involving software, electronics, vehicle dynamics, marketing and management teams.

Achievements

- 1st — Driverless Cup, Formula Student Spain 2024
- 16th Overall — Formula Student Germany 2025

Technologies

ROS, Linux, CAN communication, telemetry, embedded systems, vehicle controls, autonomous systems, simulation, validation, electric vehicle, low-pass and Kalman filters, perception and racing-line optimisation algorithms.

Research Assistant — InPercept Autonomous Driving Project

TSC UPC / Capgemini / Ficoso | Feb 2023 – May 2024

- Developed perception algorithms on NuScenes dataset for vehicle depth estimation using Sensor fusion 2D and 3D data and vehicle physical characteristics.
- Analysed the algorithm performance between camera+radar and camera+LiDAR systems.
- Exposure to machine learning technologies and sensor fusion methodologies.
- Part of the research project was conducted as part of the final degree thesis activities.

Team Leader & Software Architect — Providence+ / XPRIZE Rainforest Challenge

Feb 2023 - July 23

- Team Leader and Software Architect within a multidisciplinary 12-engineer team that supported Providence+ team participating in the XPRIZE Rainforest Challenge semi-finals.
- Responsible for planning, task coordination and tracking, and stakeholder communication.
- Delivered technical presentations and reviews to external collaborators and stakeholders.
- Providence+ team successfully advanced to the competition finals.

Education

Master's Degree in Advanced Telecommunication Technologies (MATT)

Universitat Politècnica de Catalunya (UPC) | Sept 2024 – Present

Pathway: Deep Learning for Multimedia Processing.

Bachelor's Degree in Telecommunications Technologies and Services Engineering

Universitat Politècnica de Catalunya (UPC) | Sept 2018 – June 2024

Major in Audiovisual Systems and Signal Processing.

American High School Dual Diploma Programme

Academica International Studies

Technical Skills

- **Software & Tools:** MATLAB/Simulink, Git, CANoe, Wireshark, ROS, VS Code.
- **Programming:** C, C++, Python, MATLAB, Java.
- **Embedded & Vehicle Communication:** CAN bus, LIN, TCP/IP, UDP, I2C, STM32, PIC18.
- **Operating Systems:** Linux (Ubuntu), Windows.
- **Engineering Areas:** Deep Learning, embedded systems, computer vision, vehicle controls, simulation, autonomous systems, EV powertrain, telemetry, validation, motorsport.

Personal Skills

Leadership and team management experience as Rugby Captain and engineering Team Leader.

Strong communication and stakeholder presentation skills.

Reliable, focused, proactive, and solution-oriented mindset.

Comfortable working within fast-paced multidisciplinary engineering environments.

Languages

English — Professional working proficiency

Catalan, Spanish — Native